Math **163** Name (PLEASE PRINT) Cap'n Orangebeard

Exam **2B** **2020 Fall**/Spring/Summer I /Summer II Score _____

Please fill in the following table checking boxes corresponding to parts of problems that you have completed and submitted and an expected grade. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, as per the directions of the problem. Ideally you should know exactly what you have done correctly and what grade you should receive. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1a	2	4								6	
1b	2	2	2							6	
1c											
2a	6									6	
2b	1	1	1	1	1	1				6	
2c											
3a	1	1	1	1	1	1				6	
3b	6									6	
3c											
								Total		36	

6 (1a) Find the truth-tables for:

2 (i) $\Big(p \vee (\neg q)\Big) \wedge \Big((\neg p) \vee q\Big)$

p	q	$\neg p$	$\neg q$	$p \vee (\neg q)$	$(\neg p) \vee q$	$(p \vee (\neg q)) \wedge ((\neg p) \vee q)$
0	0	1	1	1	1	1
0	1	1	0	0	1	0
1	0	0	1	1	0	0
1	1	0	0	1	1	1

4 (ii) $\Big(p \longrightarrow q\Big) \wedge \Big(q \longrightarrow r\Big)$

p	q	r	$p \longrightarrow q$	$q \longrightarrow r$	$(p \longrightarrow q) \wedge (q \longrightarrow r)$
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	0	0
0	1	1	1	1	1
1	0	0	0	1	0
1	0	1	0	1	0
1	1	0	1	0	0
1	1	1	1	1	1

6 (1b) Find truth-tables for the following expressions:

2 (i) $LS \coloneqq \left(\left(p \wedge q \right) \longrightarrow r \right)$,

2 (ii) $RS \coloneqq \left(\left(p \longrightarrow r \right) \wedge \left(q \longrightarrow r \right) \right)$, and

2 (iii) decide, if $LS \coloneqq \left(\left(p \wedge q \right) \longrightarrow r \right) \Longleftrightarrow \left(\left(p \longrightarrow r \right) \wedge \left(q \longrightarrow r \right) \right) =: RS$

p	q	r	$p \wedge q$	$(p \wedge q) \longrightarrow r$	$p \longrightarrow r$	$q \longrightarrow r$	$(p \longrightarrow r) \wedge (q \longrightarrow r)$	$LS \Longleftrightarrow RS$
0	0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	1	1
0	1	0	0	1	1	0	0	0
0	1	1	0	1	1	1	1	1
1	0	0	0	1	0	1	0	0
1	0	1	0	1	1	1	1	1
1	1	0	1	0	0	0	0	1
1	1	1	1	1	1	1	1	1

$LS \coloneqq \left(\left(p \wedge q \right) \longrightarrow r \right) \Longleftrightarrow \left(\left(p \longrightarrow r \right) \wedge \left(q \longrightarrow r \right) \right) =: RS$

See the boxed zero Y (N) (Pf) W

6 (2a) Find $\varphi(p,q,r)$ in terms of p,q,r, and connectives, if $\varphi(p,q,r)$ is to have the following truth table:

p	q	r	$\varphi(p,q,r)$	φ_1	φ_2	φ_3	φ_4	φ_5	φ_6	φ_7	φ_8
0	0	0	1	0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0	0
0	1	0	1	0	0	1	0	0	0	0	0
0	1	1	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0
1	0	1	1	0	1	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	0	0	0	0	0	0	0

$$\varphi(p,q,r)$$

$$= \varphi_1(p,q,r) \vee \varphi_2(p,q,r) \vee \varphi_3(p,q,r) \vee \varphi_4(p,q,r)$$

$$= (p \wedge q \wedge r) \vee (p \wedge (\neg q) \wedge r) \vee ((\neg p) \wedge q \wedge (\neg r)) \vee ((\neg p) \wedge (\neg q) \wedge (\neg r))$$

6 (2b) Show work on the facing side to decide, **using the algebra of propositions**, if the following hold and circle the appropriate choice:

1 (i) $\Big((\top \longrightarrow \top) \longrightarrow \bot\Big) = \top$ Y (N) (Pf) W

1 (ii) $\Big(\top \longrightarrow (\top \longrightarrow \bot)\Big) = \bot$ (Y) N (Pf) W

1 (iii) $\Big((\bot \longrightarrow \top) \longrightarrow \top\Big) = \top$ (Y) N (Pf) W

1 (iv) $\Big(\bot \longrightarrow (\top \longrightarrow \top)\Big) = \bot$ Y (N) (Pf) W

1 (v) $\Big(\top \longrightarrow (\top \longrightarrow \bot)\Big) = \top$ Y (N) (Pf) W

1 (vi) $\Big((\top \longrightarrow \top) \longrightarrow \bot\Big) = \bot$ (Y) N (Pf) W

2b.0

$$\begin{aligned} & \top \to \top \\ = \; & (\neg \top) \vee \top \\ = \; & \bot \vee \top \\ = \; & \top \end{aligned}$$

2b.1

$$\begin{aligned} & \bot \to \top \\ = \; & (\neg \bot) \vee \top \\ = \; & \top \vee \top \\ = \; & \top \end{aligned}$$

2b.2

$$\begin{aligned} & \bot \to \bot \\ = \; & (\neg \bot) \vee \bot \\ = \; & \top \vee \bot \\ = \; & \top \end{aligned}$$

2b.3

$$\begin{aligned} & \top \to \bot \\ = \; & (\neg \top) \vee \bot \\ = \; & \bot \vee \bot \\ = \; & \bot \end{aligned}$$

2b.i

$$\begin{aligned} & (\top \to \top) \to \bot \\ = \; & \top \to \bot \\ = \; & \bot \end{aligned}$$

$$\frac{\mathit{Cdn}((\top \to \top) \to \bot)}{((\top \to \top) \to \bot) \neq \top}$$

2b.ii

$$\begin{aligned} & \top \to (\top \to \bot) \\ = \; & \top \to \bot \\ = \; & \bot \end{aligned}$$

$$\frac{\mathit{Cdn}(\top \to (\top \to \bot))}{(\top \to (\top \to \bot)) = \bot}$$

2b.iii

$$\begin{aligned} & (\bot \to \top) \to \top \\ = \; & \top \to \top \\ = \; & \top \end{aligned}$$

$$\frac{\mathit{Tlg}((\bot \to \top) \to \top)}{((\bot \to \top) \to \top) = \top}$$

2b.iv

$$\begin{aligned} & \bot \to (\top \to \top) \\ = \; & \bot \to \top \\ = \; & \top \end{aligned}$$

$$\frac{\mathit{Tlg}(\bot \to (\top \to \top))}{(\bot \to (\top \to \top)) \neq \bot}$$

2b.v

$$\begin{aligned} & \top \to (\top \to \bot) \\ = \; & \top \to \bot \\ = \; & \bot \end{aligned}$$

$$\frac{\mathit{Cdn}(\top \to (\top \to \bot))}{(\top \to (\top \to \bot)) \neq \top}$$

2b.vi

$$\begin{aligned} & (\top \to \top) \to \bot \\ = \; & \top \to \bot \\ = \; & \bot \end{aligned}$$

$$\frac{\mathit{Cdn}((\top \to \top) \to \bot)}{((\top \to \top) \to \bot) = \bot}$$

6 (3a) Show work on the facing side to decide if the following proposition, denoted by ψ is a tautology, contradiction, or contingency, and circle an appropriate answer.

1 (i) $\left((\top\longrightarrow\top)\longrightarrow\bot\right)$ $Tlg(\psi)$ $Cdn(\psi)$ $Cng(\psi)$

See page (5080): 2b. i

1 (ii) $\left(\top\longrightarrow(\top\longrightarrow\bot)\right)$ $Tlg(\psi)$ $Cdn(\psi)$ $Cng(\psi)$

See page (5080): 2b. ii

1 (iii) $\left((\bot\longrightarrow\top)\longrightarrow\top\right)$ $Tlg(\psi)$ $Cdn(\psi)$ $Cng(\psi)$

See page (5080): 2b. iii

1 (iv) $\left(\bot\longrightarrow(\top\longrightarrow\top)\right)$ $Tlg(\psi)$ $Cdn(\psi)$ $Cng(\psi)$

See page (5080): 2b. iv

1 (v) $\left(\top\longrightarrow(\top\longrightarrow\bot)\right)$ $Tlg(\psi)$ $Cdn(\psi)$ $Cng(\psi)$

See page (5080): 2b. v

1 (vi) $\left((\top\longrightarrow\top)\longrightarrow\bot\right)$ $Tlg(\psi)$ $Cdn(\psi)$ $Cng(\psi)$

See page (5080): 2b. vi

6 (3b) Show work on the facing side to decide, **using axioms and rules of inference**, if the following arguments are valid and circle the appropriate choice:

$$Vld \left(\frac{\top \rightarrow \left(p \wedge q\right) \quad \left(p \vee q\right) \rightarrow \top}{\left(p \vee q\right) \rightarrow \left(p \wedge q\right)} \right) \quad (Y) \quad N \quad (Pf) \quad W$$

3b

$$\dfrac{\dfrac{\dfrac{}{\mathrm{Vld}\left(\dfrac{p\to q \quad q\to r}{p\to r}\right)}}{\mathrm{Vld}\left(\dfrac{q\to r \quad p\to q}{p\to r}\right)}}{\mathrm{Vld}\left(\dfrac{\top\to(p\wedge q) \quad (p\vee q)\to\top}{(p\vee q)\to(p\wedge q)}\right)}\left\langle\begin{array}{l} p\leftarrow(p\vee q)\\ q\leftarrow\top\\ r\leftarrow(p\wedge q)\end{array}\right\rangle$$